LBS COLLEGE OF ENGINEERING KASARAGOD
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

MOCKMATE AI:

An AI-Powered Interview Platform for Students



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PROJECT PHASE I | Academic Year 2026-27

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INTRODUCTION

- Technical interviews test a candidate's knowledge, problem-solving ability and communication.
- Students and freshers get few chances to practise in a realistic setting; human mock interviews are not always accessible.
- MockMate AI is a web-based interview-practice platform for students.
- **Phase I (current):** the user selects a topic and difficulty, answers five questions by typing or recording audio, and receives a scored report with strengths, weaknesses and study suggestions.
- Camera-based behaviour analysis and adaptive questioning are **planned** for later phases.

> **Main idea:** give students a simple way to practise interviews, see their weak areas, and improve before a real interview.

Literature Survey (1/4)

Sl. No.	Title	Author	Methodology	Advantage	Disadvantage
1	AI Mock Interview System	Madhura Abhishek; Meenakshi Sheik	Generates interview questions, records answers, analyses responses, and provides feedback.	Convenient practice; quick feedback; no human interviewer needed.	May miss some human communication details; feedback depends on system quality.
2	Rider-AI Powered Mock Interview System with Real-Time Voice and Emotion Analysis	Soni Khapekar et al.	Simulated interview with question generation, voice analysis, and emotion analysis.	Interactive practice; combines voice and emotion analysis.	Emotion detection may be inaccurate; recording quality affects results.
3	FairHire: AI-Driven Resume Profiling and Technical Interview Automation	Akshay Baby et al.	Extracts resume skills, builds candidate profiles, and generates technical questions.	Speeds up screening; links resume analysis with interviews.	Depends on candidate data quality; response understanding can be difficult.

Literature Survey (2/4)

Sl. No.	Title	Author	Methodology	Advantage	Disadvantage
4	AI Based Mock Interview System Using NLP (2025)	K. Senthilkumar et al.	Uses STT, NLP, reinforcement learning, OpenCV, MediaPipe, and TTS.	Covers technical, communication, and behavioural aspects.	Noise and lighting can affect results; several technologies add complexity.
5	MAG-BERT-ARL for Fair Automated Video Interview Assessment (2024)	Bimasena Putra et al.	Uses BERT with audio and visual information, multimodal adaptation, and adversarial learning.	Combines multiple input types and considers fairness.	Complex model; needs computing power and suitable training data.
6	Candidate Evaluation with Multimodal Data-Driven for Recruitment (2024)	Xing Wu et al.	Uses GNN/GAT and DP-GNN to process text, audio, and image information.	Combines job matching with multimodal evaluation.	Needs high-quality data; personality prediction may be inaccurate; not focused on adaptive technical questions.

Literature Survey (3/4)

Sl. No.	Title	Author	Methodology	Advantage	Disadvantage
7	Conversate: Supporting Reflective Learning in Interview Practice Through Interactive Simulation and Dialogic Feedback (2025)	Taufiq Daryanto et al.	Whisper for speech-to-text; GPT-3.5-Turbo for simulation; GPT-4 for feedback; MongoDB and Firebase storage.	Adaptive follow-up questions and interactive feedback.	Quality and consistency depend on AI-generated questions and feedback.
8	MOCKLLM: A Multi-Agent Behavior Collaboration Framework for Online Job Seeking and Recruiting (2025)	Hongda Sun et al.	Nanbeige-16B-Chat, multi-agent role-play, vLLM, reflection memory, and handshake protocol.	Supports multi-turn practice and uses reflection to improve later responses.	More complex and resource-intensive than single-agent systems.
9	Single- vs. Dual-Prompt Dialogue Generation with LLMs for Job Interviews (2025)	Joachim De Baer et al.	GPT-4o and Llama 3.3 70B; interviewer/candidate role agents; Llama judging via Groq.	Can create more human-like dialogue; role-based interaction.	Dual prompts use more tokens and increase API/computing cost.

Literature Survey (4/4)

Sl. No.	Title	Author	Methodology	Advantage	Disadvantage
10	Candidate Evaluation with Multimodal Data-Driven for Recruitment (ICPR 2024)	X. Wu et al.	Combines graph neural networks with text, audio, and image processing.	Supports multimodal candidate assessment.	Requires quality data and does not focus on adaptive technical questions.
11	Multi-Label Multimodal Emotion Recognition with Transformer-Based Fusion (2023)	Y. Wang et al.	Processes video, audio, and text using transformer-based multimodal fusion.	Recognises multiple emotions and relationships between input types.	Computationally demanding; mainly focuses on emotion recognition.
12	Experience-Aware Question Generation: A Personalized Approach (2025)	Mallikarjuna G. D. et al.	Analyses resume and experience using NLP and Bloom's Taxonomy to set question difficulty.	Personalised questions that match experience and difficulty.	Depends on resume details; question accuracy can vary; no video analysis.

GAPS IDENTIFIED

- Many systems simulate interviews, but the depth of technical feedback can be limited.
- Some analyse voice or emotion without properly assessing technical answers.
- Resume-based question generation may not adapt to the candidate’s actual answers.
- Multimodal systems target recruitment or emotion recognition rather than student practice.
- AI-generated questions and feedback can be inaccurate or inconsistent.

Identified gap

A student-focused platform for realistic practice with understandable technical feedback in one place.

Our response

IMPLEMENTED Concept-based technical feedback

IMPLEMENTED Typed and spoken answers

PLANNED Adaptive questions

PLANNED Camera / behaviour analysis

MOTIVATION

- Help students and freshers practise interviews before applying for jobs.
- Reduce nervousness through repeated, low-pressure practice.
- Make practice accessible without depending on another person.
- Show students their strengths, missing concepts and topics to revise.
- Start with technical question practice and keep the platform open to other domains, and later to communication and behaviour feedback.



STAKEHOLDER SURVEY

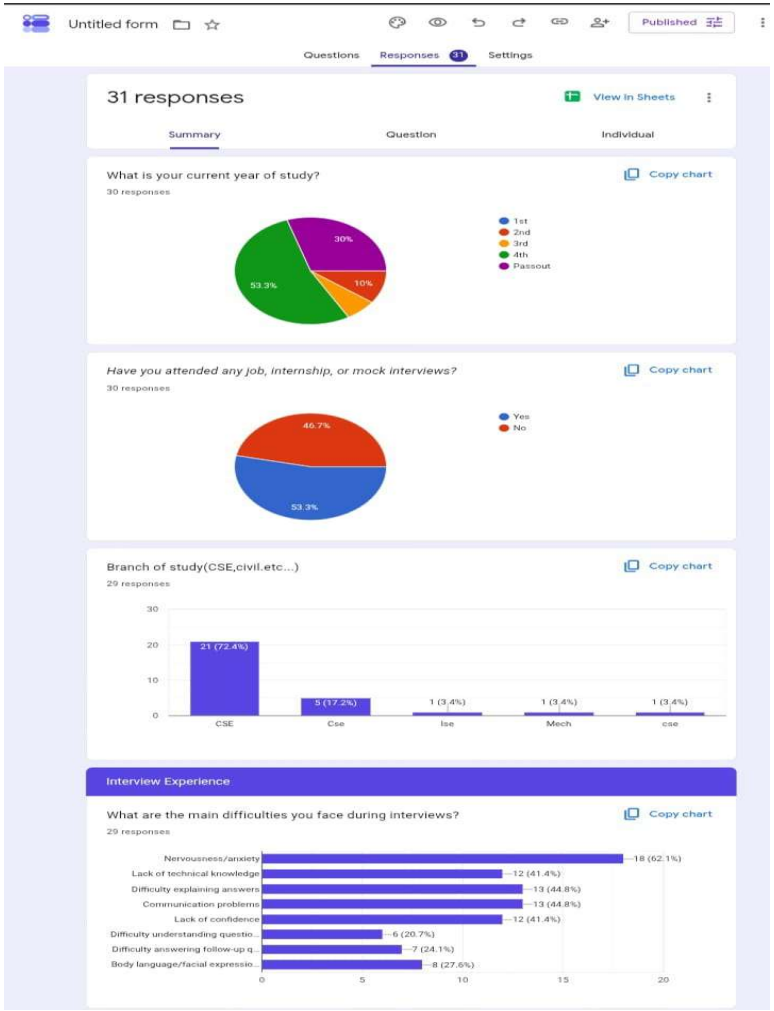
Students

- May feel nervous or lack confidence in interviews.
- Need realistic practice and quick feedback.
- Need pointers on weak areas to work on.
- Need questions that suit their level (easy / medium / hard).

Recruiters

- Look for technical knowledge and clear communication.
- Evaluating many candidates manually takes time.
- Consistent assessment supports fairer decisions.
- Clear reports help identify strengths and weaknesses.

Stakeholder needs are drawn from the literature survey. The platform is aimed mainly at **students**; recruiter-side use is outside the Phase I scope.


Untitled form
Published
Questions
Responses 31
Settings
31 responses
View in Sheets
Summary
Question
Individual
What is your current year of study?
30 responses
Copy chart
30%
10%
53.3%
1st
2nd
3rd
4th
Passout
Have you attended any job, internship, or mock interviews?
30 responses
Copy chart
46.7%
53.3%
Yes
No
Branch of study(CSE,civil.etc...)
29 responses
Copy chart
21 (72.4%)
5 (17.2%)
1 (3.4%)
1 (3.4%)
1 (3.4%)
CSE
Cse
Ise
Mech
cse
Interview Experience
What are the main difficulties you face during interviews?
29 responses
Copy chart
Nervousness/anxiety
18 (62.1%)
Lack of technical knowledge
12 (41.4%)
Difficulty explaining answers
13 (44.8%)
Communication problems
13 (44.8%)
Lack of confidence
12 (41.4%)
Difficulty understanding questio...
6 (20.7%)
Difficulty answering follow-up q...
7 (24.1%)
Body language/facial expressio...
8 (27.6%)


Untitled form
Published
Questions
Responses 31
Settings
Would you like the AI to ask follow-up questions based on your previous answers?
30 responses
Copy chart
10%
83.3%
Yes
No
Maybe
Would you prefer interview questions customized according to your resume/job role?
30 responses
Copy chart
93.3%
Yes
No
Maybe
How useful would feedback about your performance be after the interview?
30 responses
Copy chart
12 (40%)
7 (23.3%)
5 (16.7%)
3 (10%)
3 (10%)
What type of feedback would you like to receive?
30 responses
Copy chart
Strength
23 (76.7%)
Weekness
21 (70%)
Technical performance
22 (73.3%)
Confidence
21 (70%)
Facial expressions/body langua...
15 (50%)
Suggestions for improvement
25 (83.3%)
Overall score
20 (66.7%)

PROBLEM STATEMENT

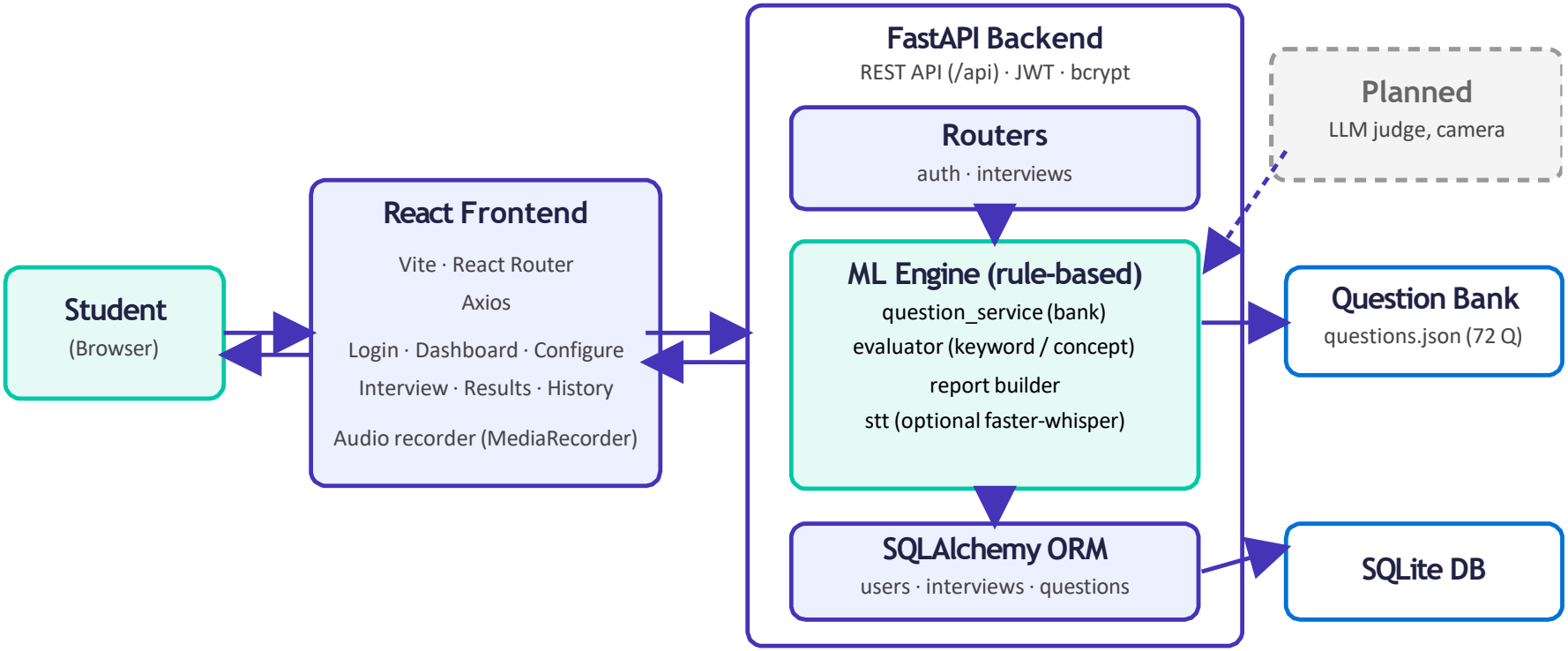
To develop **"MockMate AI"**, a user-friendly web application for KTU students that conducts technical mock interviews, accepts typed or spoken answers, evaluates the technical content of each answer, and generates personalised feedback on strengths, missing concepts and topics to revise.

Phase I scope: question bank, answer evaluation, report generation, user accounts and web interface.
Planned: camera-based behaviour analysis, communication and confidence assessment, adaptive questions.

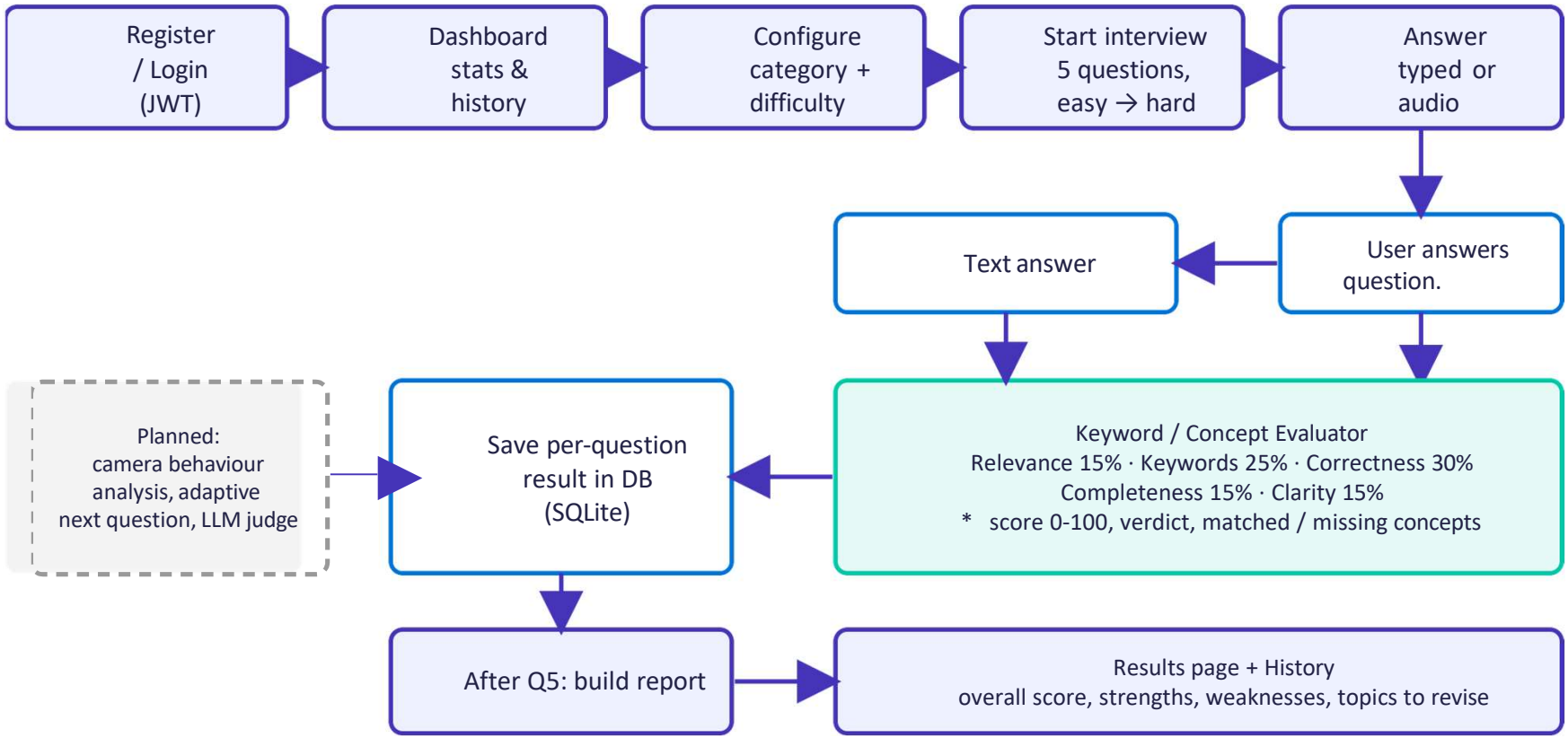
OBJECTIVES

IMPLEMENTED	Develop an interview-practice web platform for students
IMPLEMENTED	Begin with Python technical questions (bank also covers 8 other categories)
PARTIAL	Accept answers by chat (typed) and audio; transcription needs an optional engine
PLANNED	Accept camera-based input for behaviour analysis
IMPLEMENTED	Analyse technical answers (concept coverage, completeness, clarity)
PLANNED	Assess communication, confidence and observable behaviour
IMPLEMENTED	Generate personalised feedback: strengths, weaknesses, topics to revise
PLANNED	Support more domains, adaptive questions, resume personalisation

SYSTEM ARCHITECTURE



DETAILED BLOCK DESIGN



IMPLEMENTATION ANALYSIS

MODULE	TECHNOLOGY / DETAILS	STATUS
Backend API	FastAPI, Uvicorn, Pydantic	IMPLEMENTED
Authentication	JWT (PyJWT), bcrypt password hashing	IMPLEMENTED
Database	SQLAlchemy 2 + SQLite: users, interviews, interview_questions	IMPLEMENTED
Question bank	JSON, 9 categories × 8 questions = 72, with key concepts	IMPLEMENTED
Evaluation	Rule-based keyword / concept scorer, 5 weighted dimensions	IMPLEMENTED
Report builder	Rule-based strengths, weaknesses, missing concepts, suggestions	IMPLEMENTED
Frontend	React 18, Vite, React Router, Axios — 8 pages	IMPLEMENTED
Audio answers	Browser MediaRecorder; server transcription via optional faster-whisper	PARTIAL
Testing / CI	pytest (10 tests defined), GitHub Actions workflow	PARTIAL

IMPLEMENTATION ANALYSIS: STATUS

Completed

- Register / login with JWT; owner-only access to interviews
- Interview lifecycle: start, answer, resume, complete
- Selection of 5 questions, ordered easy → hard
- Keyword / concept evaluator and score breakdown
- Report with classification and suggestions
- Dashboard, history and results pages

Partially implemented

- Audio: recording and upload work; evaluation only when faster-whisper is installed (otherwise API returns 501)
- Evaluator checks concept coverage, not factual truth
- Extensible interfaces for an LLM evaluator and question source exist; no LLM is connected

Not yet implemented

- Camera / video analysis
- Confidence and communication scoring
- Adaptive follow-up questions
- Resume-based personalisation
- Rate limiting, email verification

PROJECT PLAN

PHASE I — INITIAL DEVELOPMENT (CURRENT)

REQUIREMENTS & DESIGN	BACKEND	EVALUATION ENGINE	FRONTEND & TESTING
■ Literature survey ■ Problem definition ■ System architecture	■ FastAPI + SQLite models ■ JWT authentication ■ Interview API	■ Question bank (72) ■ Keyword evaluator ■ Report builder ■ STT adapter	■ React pages ■ Audio recorder ■ pytest suite and CI

PHASE II — PLANNED

ANSWER EVALUATION	QUESTIONS	CAMERA / BEHAVIOUR	VALIDATION
■ Add LLM / NLP evaluator ■ Compare with keyword scores	■ Expand bank ■ Adaptive follow-ups ■ Resume based question.	■ Capture webcam input ■ Observable-behaviour features	■ Trial with students ■ Fix issues, harden security

RESULT

Output of the implemented evaluator on the question *"What is the difference between a list and a tuple in Python?"* (8 key concepts):

ANSWER	SUMMARY OF ANSWER	SCORE	VERDICT	CONCEPTS
Complete	Mutable vs immutable, brackets, hashable, performance	**100**	Correct	8 of 8 matched
Short	"A list can be changed but a tuple cannot be changed."	**70**	Correct	4 of 8 matched; missing: square brackets, parentheses, hashable, faster
Off-topic	"I like cricket and playing football on weekends."	**10**	Incorrect	0 of 8 matched

Question selection (Python, easy):
1. Lists vs tuples 2. Equality vs identity 3. Dictionaries 4. Decorators 5. Generators
3 easy + 2 medium, ordered easy → hard

Observation: a short answer with only part of the concepts still scored 70. The evaluator rewards key terms and cannot verify correctness — a known limitation to address in Phase II.

RESULT: IMPLEMENTED INTERFACES

PAGE	PURPOSE
Landing	Introduction and entry point
Register / Login	Account creation, JWT sign-in
Dashboard	Interview statistics, start new interview
Configure	Choose category and difficulty
Interview	Question-by-question answering, typed or recorded
Results	Score ring, breakdown, report
History	Past interviews; resume unfinished ones

METHOD	ENDPOINT
POST	/api/auth/register, /api/auth/login
GET	/api/auth/me, /api/meta, /api/stats
POST / GET	/api/interviews
GET	/api/interviews/{id}
POST	/api/interviews/{id}/answer
POST	/api/interviews/{id}/answer/audio
GET	/api/interviews/{id}/report

CONCLUSION

- MockMate AI is in its initial development phase. A working web application is in place for text and audio-based technical interview practice.
- The current system supports registration, topic and difficulty selection, five-question interviews, keyword-based scoring and a personalised report.
- Camera-based behaviour analysis, adaptive questions and LLM-based evaluation are not yet implemented and form the next phase.
- The modular design allows these components to be added without restructuring the application.

FUTURE WORK

- Replace or complement keyword scoring with an LLM / NLP-based evaluator.
- Adaptive follow-up questions based on the candidate’s answers.
- Camera input for observable behaviour; communication and confidence feedback.
- Larger question bank, more programming languages and technical domains.
- Resume-based personalisation.
- Security hardening: httpOnly cookies, rate limiting, email verification.
- Trial with students and evaluation of feedback quality.

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1. Madhura Abhishek and Meenakshi Sheik, “AI Mock Interview System.”
2. Soni Khapekar et al., “Rider-AI Powered Mock Interview System with Real-Time Voice and Emotion Analysis.”
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4. K. Senthilkumar et al., “AI Based Mock Interview System Using NLP,” 2025.
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12. Mallikarjuna G. D. et al., “Experience-Aware Question Generation: A Personalized Approach,” 2025.

THANK YOU